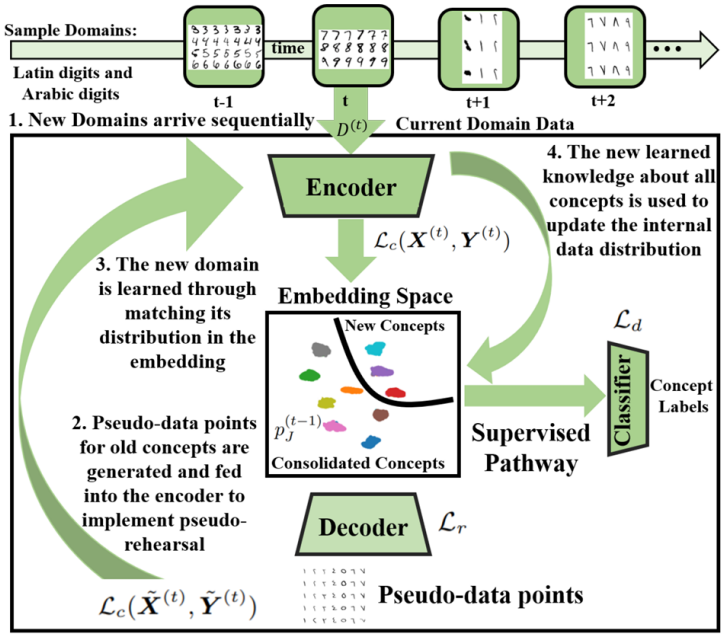
COMSPCI 713 Part 2 – Mock questions

Question 1 [4 marks]

The paper *"Cognitively Inspired Learning of Incremental Drifting Concepts"* by Rostami & Galstyan introduces a continual learning system designed to handle concept drift. The diagram below illustrates the core steps of this method:



Incremental Concept Learning Process

Using your knowledge of continual learning from the lecture, answer the following questions:

1) What specific challenge in continual learning is this approach designed to address? [1 mark]
2) Step 2 in the diagram shows the generation of pseudo-data points. Which of the three major categories of continual learning strategies covered in class does this step belong to? Briefly describe the key idea behind this category. [2 marks]
3) The approach uses a Gaussian Mixture Model (GMM). What is the purpose of the GMM in this system? [1 mark]

Question 2 [3 marks]

Uniform Cost Search (UCS) and Breadth-First Search (BFS) are two commonly used tree search algorithms.

1) Describe how each algorithm explores a search tree. [1 mark]
2) Suppose the edges in a search tree have different costs. For each algorithm, state whether it is optimal in this case and explain why. [2 marks]

Question 3 [3 marks]

Monte Carlo Tree Search (MCTS) can be implemented using an Upper Confidence Tree (UCT), which relies on the Upper Confidence Bound (UCB1) formula shown below:

$$UCB1(n) = \frac{U(n)}{N(n)} + C \times \sqrt{\frac{\log N(\text{PARENT}(n))}{N(n)}}$$

Where:

- $U(n)$: total utility (reward) from simulations passing through node n
- $N(n)$: number of times node n has been visited

Briefly explain the role of each of the following components in the formula:

1. $\frac{U(n)}{N(n)}$
2. C
3. $\sqrt{\frac{log(N(PARENT(n))}{N(n)}}$

Question 4 [1 mark]

In a simplified Pac-Man environment, the agent must eat all dots in a maze as efficiently as possible, while avoiding moving ghosts. Pac-Man can move in four directions and sees nearby tiles. The goal is to clear the maze efficiently without getting caught by the ghosts.

Which option best defines the **State space**, **Action space**, **Policy**, and **Reward** of a Reinforcement Learning scenario? [1 mark for selecting the correct answer, no explanation requires]

A)

- **State space:** Pac-Man's position, nearby dot positions and ghost positions
- **Action space:** {up, down, left, right}
- **Policy:** Mapping of Pacman's position to action probabilities
- **Reward:** +10 for eating a dot, -100 if caught by a ghost, -1 per move

B)

- **State space:** Pac-Man's position, nearby dot positions and ghost positions
- **Action space:** {up, down, left, right}
- **Policy:** Mapping of the observed state to action probabilities
- **Reward:** +10 for eating a dot, -1 per move

C)

- **State space:** Pac-Man's position, nearby dot positions and ghost positions

- **Action space:** {up, down, left, right}
- **Policy:** Mapping of the observed state to action probabilities
- **Reward:** +10 for eating a dot, -100 if caught by a ghost, -1 per move

D)

- **State space:** Pac-Man's position only
- **Action space:** {up, down, left, right}
- **Policy:** Deterministic path from start to finish
- **Reward:** +1 per move, +100 when all dots eaten

Question 5 [2 marks]

In the context of Graph Neural Networks (GNNs), briefly explain the concepts of permutation invariance and permutation equivariance.

Question 6 [2 marks]

In the context of Self-Supervised Learning (SSL):

1. What is the difference between a pretext task and a downstream task? Provide a brief explanation. [1 mark]
2. When evaluating a self-supervised learning system, which task's performance is more important, and why? [1 mark]